Lecture 8

Factoring

Objectives. *Lecture 7 gave a circuit for estimating eigenphases. This lecture asks:* how does a phase become a factor? *A gcd reduces factoring to the order r of a random a modulo N (Proposition 8.2). Starting in $|1\rangle$, phase estimation for multiplication by a estimates s/r. Continued fractions and repeated verification recover r with bounded error (Theorem 8.9). This factors an n-bit N using $O(n^3)$-gate circuits (Theorem 8.4); no classical polynomial-time algorithm is known.*

By the end of this lecture you will be able to:

- Prove that $\gcd(a^{r/2}-1, N)$ is a proper factor of N when r is even and $a^{r/2} \not\equiv -1$, and state how often a uniformly chosen unit qualifies for odd N with at least two distinct prime factors.
- Prove that $|1\rangle$ is the uniform superposition of the r orbit eigenvectors of U_a, and deduce its phase-estimation distribution.
- Prove that $3n$ runs with $t = 2n+1$ counting qubits, continued fractions and a classical test find r with probability at least $2/3$.
- State factoring's proved, cited and conjectured claims.

8.1 The reduction, its gcd, preprocessing and probability

Factoring asks, for a composite N, for a *proper factor*: a divisor d of N with $1 < d < N$. In this lecture N is the number to factor, no longer 2^n as in Lecture 7; it has n bits. Two cases are classical and fast: an even N has the factor 2, and a power $N = b^k$, $k \geq 2$, the factor b, the integer k-th root of N for some $k < n$ (Nielsen and Chuang, Ex. 5.17). Since N is composite, what remains is an odd N with at least two distinct prime factors.

Choose a uniformly in $\{1, \ldots, N-1\}$. Euclid's algorithm gives $\gcd(a, N)$ in time polynomial in n; if it is not 1, it is a proper factor. Extended, it gives integers u and v with $ua + vN = \gcd(a, N)$ (Kitaev, Shen and Vyalyi, p. 40). So an a coprime to N, a *unit* modulo N, has an *inverse* u: $ua \equiv 1$.

Definition 8.1 (The order)**.** For a coprime to N, the *order* of a modulo N is the least $r \geq 1$ with $a^r \equiv 1 \pmod{N}$.

It exists, and $r < N$: the powers $a^0, \ldots, a^{N-1}$ are units, of which there are at most $N-1$, so two agree, $a^i \equiv a^j$ with $i < j$, and multiplying by the inverse of a^i gives $a^{j-i} \equiv 1$. (This r is not the Bloch vector of QM-I, Lecture 3.)

Proposition 8.2 (The reduction · *proved modulo extended Euclid*)**.** *Let a be coprime to N, with order r. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2}-1, N)$ and $\gcd(a^{r/2}+1, N)$ are proper factors of N.*

Proof. Let $b = a^{r/2} \bmod N$. Then N divides $b^2 - 1 = (b-1)(b+1)$. It divides neither factor: $b \not\equiv 1$, as $r/2 < r$ and r is the least, and $b \not\equiv -1$ by hypothesis. If $\gcd(b-1, N)$ were 1, multiplying $(b-1)(b+1) \equiv 0$ by the inverse of $b-1$ would give $b+1 \equiv 0$; if it were N, N would divide $b-1$. So $1 < \gcd(b-1, N) < N$, and likewise for $b+1$. ∎

Theorem 8.3 (A random base · *cited*)**.** *Let N be odd with $m \geq 2$ distinct prime factors, and let a be uniform among the units modulo N. Then r is even and $a^{r/2} \not\equiv -1$ with probability at least $1 - 2^{-(m-1)} \geq \frac{1}{2}$.*

An a uniform in $\{1, \ldots, N-1\}$, once it is coprime to N, is uniform among the units. It is Kitaev, Shen and Vyalyi's (pp. 121–122) and Shor's (p. 16). Nielsen and Chuang's Theorem 5.3 (p. 233) prints $1 - 2^{-m}$, which is false: for $N = 21$ the probability is exactly $\frac{1}{2}$.

8.2 U_a, its eigenvectors, $|1\rangle$ as their superposition

The whole algorithm, Shor's, is the following. Its one quantum step, order finding, takes this section and the next two; the proof ends in §8.5.

Theorem 8.4 (Factoring · *proved modulo the cited lemmas*)**.** *For a composite n-bit N, the following finds a proper factor with probability at least $2/3$.*

1. *Return 2 if N is even, and b if $N = b^k$, $k \geq 2$.*
2. *Otherwise, make at most three trials. In each trial:*
   (a) *Draw at most seven uniform n-bit strings. Let a be the first with $1 \leq a < N$; if there is none, skip this trial.*
   (b) *Return $\gcd(a, N)$ if it is not 1.*
   (c) *Otherwise run Theorem 8.9. If it returns an even r, return $\gcd(a^{r/2} - 1, N)$ if that is a proper factor.*

It uses at most $9n$ circuits of $O(n^3)$ gates over the alphabet, and classical work polynomial in n.

The draws make a uniform in $\{1, \ldots, N-1\}$ in bounded time: an odd N with n bits has $N - 1 \geq 2^{n-1}$, so a draw qualifies with probability at least $\frac{1}{2}$, and seven find a with probability at least $1 - 2^{-7}$.

For a coprime to $N < 2^n$, define the gate U_a on the n-qubit register by

$$U_a |y\rangle = \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases}$$

As a has an inverse modulo N, the map $y \mapsto ay \bmod N$ permutes $\{0, \ldots, N-1\}$, so U_a permutes the basis states and is unitary. On $|1\rangle$ it walks the orbit

$$|1\rangle \mapsto |a\rangle \mapsto \left|a^2\right\rangle \mapsto \cdots \mapsto \left|a^{r-1}\right\rangle \mapsto |1\rangle,$$

with the powers taken modulo N: r distinct basis states, which U_a shifts round a cycle of length r. Its eigenvectors there are built from the characters of $\mathbb{Z}_r$, as the columns of Lecture 7's QFT were from those of $\mathbb{Z}_N$.

Lemma 8.5 (The eigenvectors of U_a). *For $s = 0, \dots, r-1$ let*

$$|u_s\rangle = r^{-1/2} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} \left|a^k\right\rangle .$$

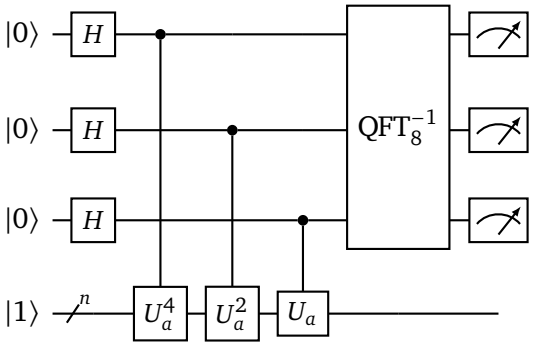
(i) $U_a |u_s\rangle = e^{2\pi i s/r} |u_s\rangle$.
(ii) The $|u_s\rangle$ are orthonormal.
(iii) $r^{-1/2} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$.

Proof. (i) U_a maps $\left|a^k\right\rangle$ to $\left|a^{k+1}\right\rangle$, and $|a^r\rangle = \left|a^0\right\rangle$, so it shifts the index k by 1 modulo r. As $e^{-2\pi i s k/r}$ has the period r in k,

$$\begin{aligned} U_a |u_s\rangle &= r^{-1/2} \textstyle\sum_k e^{-2\pi i s(k-1)/r} \left|a^k\right\rangle \\ &= e^{2\pi i s/r} |u_s\rangle . \end{aligned}$$

(ii) The $\left|a^k\right\rangle$, $k < r$, are orthonormal, so $\langle u_s | u_{s'}\rangle = r^{-1} \sum_k e^{2\pi i (s-s')k/r}$, which the geometric sum of Lecture 7 makes 1 if $s = s'$ and 0 otherwise. (iii) The sum is $r^{-1} \sum_k \left(\sum_s e^{-2\pi i s k/r}\right) \left|a^k\right\rangle$, and the inner sum is r for $k = 0$ and 0 for $0 < k < r$. ∎

We cannot prepare $|u_s\rangle$, which needs r. But $|1\rangle$ needs only an X on the bottom wire, and by (iii) it is the uniform superposition of the $|u_s\rangle$. So run phase estimation with $U = U_a$ on $|1\rangle$; for $t = 3$,



By Theorem 7.6, whose proof uses only that the $|u_s\rangle$ are orthonormal eigenvectors, the reading y has the probability $r^{-1} \sum_s p_{s/r}(y)$, with $p_{s/r}$ the distribution of Theorem 7.4 for the phase s/r. The circuit acts as if s were drawn uniformly from $\{0, \dots, r-1\}$ and s/r then estimated. The target is never read.

Two guards of earlier lectures bite on the controlled powers. They must control U_a itself, exactly, not a gate equal to it up to a global phase (Example 6.8). And a workspace they use must end in $|0 \cdots 0\rangle$: left depending on y, it would tell the branches apart, as garbage did (Example 5.2).

8.3 Precision $t = 2n + 1$ and continued fractions

Fix s. Call the reading *good* if it is within 2^{-t-1} of s/r in the circular distance. By Theorem 7.4 it is good with probability at least $4/\pi^2$, and at least $4/\pi^2 - \delta$ when $\mathrm{QFT}_{2^t}^{-1}$ is compiled *within* δ, moving no probability by more than δ. Take $t = 2n + 1$. On a good reading the distance is the plain one: for $s = 0$ the phase 0 is read exactly, as $y = 0$; for $s \neq 0$, s/r is at least $1/r > 2^{-t-1}$ from 0 and from 1, so no good reading wraps round. And as $r < N < 2^n$,

$$\left| \frac{y}{2^t} - \frac{s}{r} \right| \leq 2^{-2n-2} < \frac{1}{2r^2}.$$

A rational x has a continued fraction

$$x = c_0 + \cfrac{1}{c_1 + \cfrac{1}{\ddots + \cfrac{1}{c_m}}},$$

with integers $c_0 = \lfloor x \rfloor$ and $c_1, \ldots, c_m \geq 1$, found by Euclid's algorithm on its numerator and denominator. Its *convergents* are the fractions obtained by stopping at $c_0, c_1, \ldots, c_m$ in turn. For $x = y/2^t$ there are $O(t)$ of them, as Euclid's algorithm at least halves the remainder every two steps.

Theorem 8.6 (Continued fractions · *cited*). *Let x be rational and p/q a fraction in lowest terms, $q \geq 1$, with $|x - p/q| < 1/(2q^2)$. Then p/q is a convergent of x.*

This is Nielsen and Chuang's Theorem 5.1 (p. 229, proof pp. 637–638), and Kitaev, Shen and Vyalyi's Theorem A.13 (p. 250).[1]

On a good reading, write s/r in lowest terms as s'/r', with r' a divisor of r. Then $|y/2^t - s'/r'| < 1/(2r^2) \leq 1/(2r'^2)$, so s'/r' is a convergent of $y/2^t$, with $r' \leq r < N$. A run therefore tests the denominators $q < N$ of the convergents of $y/2^t$; r' is among them, and $r' = r$ exactly when $\gcd(s, r) = 1$.

For example, take $N = 21$ and $a = 2$. The orbit of 1 is 1, 2, 4, 8, 16, 11, so its order is 6. Here $n = 5$ and $t = 11$. Suppose the register reads $y = 341$. To process that reading, divide:

$$\begin{aligned} 2048 &= 6 \cdot 341 + 2, \\ 341 &= 170 \cdot 2 + 1, \\ 2 &= 2 \cdot 1 + 0. \end{aligned}$$

The leading coefficient is $c_0 = 0$ since $341 < 2048$. The quotients then give $341/2048 = [0; 6, 170, 2]$. The convergents can be computed without nested divisions by the recurrence

$$\begin{aligned} p_{-2} &= 0, \quad p_{-1} = 1, \\ q_{-2} &= 1, \quad q_{-1} = 0, \\ p_j &= c_j p_{j-1} + p_{j-2}, \\ q_j &= c_j q_{j-1} + q_{j-2}. \end{aligned}$$

j	c_j	p_j	q_j	p_j/q_j
0	0	0	1	0/1
1	6	1	6	1/6
2	170	170	1021	170/1021
3	2	341	2048	341/2048

For example, $p_2 = 170 \cdot 1 + 0 = 170$ and $q_2 = 170 \cdot 6 + 1 = 1021$. Only the denominators 1 and 6 are below 21. The modular test rejects 1 and accepts 6, since $2^1 \not\equiv 1$ and $2^6 = 64 \equiv 1 \pmod{21}$. Then $2^3 = 8 \not\equiv -1$, and $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$. The analysis calls this a good reading for $s = 1$, since $2^{11}/6 \approx 341.3$. The postprocessing itself is given only N, a and y; it is not given s or the order.

[1] Nielsen and Chuang print $\leq$; for the expansion Euclid's algorithm gives, that fails at $x = \frac{1}{2}$ and $p/q = \frac{1}{1}$, as the convergents of $\frac{1}{2}$ are $\frac{0}{1}$ and $\frac{1}{2}$.

8.4 Coprimality, verification, repetitions

A run finds r itself when its reading is good and $\gcd(s, r) = 1$. How likely is the second, and how is r recognised?

Proposition 8.7 (Coprimality · *proved modulo Euler's product*)**.** *Let $r \geq 1$ have k distinct prime factors, and let s be uniform in $\{0, \dots, r-1\}$. Then*

$$\begin{aligned} \Pr\big[\gcd(s, r) = 1\big] &= \left|\mathbb{Z}_r^\times\right| / r \\ &= \textstyle\prod_{p \mid r}(1 - 1/p) \\ &\geq 1/(k+1). \end{aligned}$$

If $r < 2^n$, the probability is at least $1/n$.

Proof. $\left|\mathbb{Z}_r^\times\right|$ counts the $s < r$ coprime to r, and the product is Euler's (Nielsen and Chuang, (A4.23), p. 631). List the primes dividing r as $p_1 < \cdots < p_k$. The i-th prime number is at least $i+1$, so $p_i \geq i+1$ and the product is at least $\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{k}{k+1} = \frac{1}{k+1}$. Finally $r \geq p_1 \cdots p_k \geq 2^k$, so $r < 2^n$ gives $k + 1 \leq n$. ∎

Lemma 8.8 (Verification)**.** *If $a^d \equiv 1 \pmod N$ with $d \geq 1$, then r divides d.*

Proof. Write $d = qr + \rho$ with $0 \leq \rho < r$. Then $1 \equiv a^d = (a^r)^q a^\rho \equiv a^\rho$, and $\rho \geq 1$ would contradict the minimality of r. ∎

So the test $a^q \equiv 1 \pmod N$, by repeated squaring, passes r and its multiples, and none of its proper divisors. In the example, $s = 2$ gives $\frac{1}{3}$, and $2^3 \not\equiv 1 \pmod{21}$. A bad reading can pass a multiple of r: a passing q shows only that r divides it, so we keep the least.

Theorem 8.9 (Order finding · *proved modulo continued fractions and Euler's product*)**.** *Let a be coprime to $N < 2^n$, with order r.*

1. *Run phase estimation of U_a on $|1\rangle$ with $t = 2n+1$ counting qubits and $\mathrm{QFT}_{2^t}^{-1}$ compiled within $\delta = 1/200$, for $3n$ runs.*
2. *For each reading y, test every denominator $q < N$ of a convergent of $y/2^t$. Retain q if $a^q \equiv 1 \pmod N$.*
3. *After all $3n$ runs, return the least retained denominator, or nothing if none was retained.*

The return is r with probability at least $2/3$.

Proof. For each s the reading is good with probability at least $4/\pi^2 - \delta \geq 2/5$, and s is uniform, so by Proposition 8.7 a run passes r with probability at least $2/(5n)$. By Lemma 8.8 every passing q is a multiple of r, so the least is r once some run passes r. All $3n$ runs miss with probability at most $(1 - 2/(5n))^{3n} \leq e^{-6/5} < 1/3$. ∎

8.5 Modular powers

Phase estimation needs $\mathrm{C}(U_a^{2^j})$ for $j < t$. From $\mathrm{C}(U_a)$ that is $2^t - 1$ uses (Lecture 7). But $U_a^{2^j} = U_{b_j}$ with $b_j = a^{2^j} \bmod N$, computed classically by j squarings: one controlled multiplication by a known constant each.

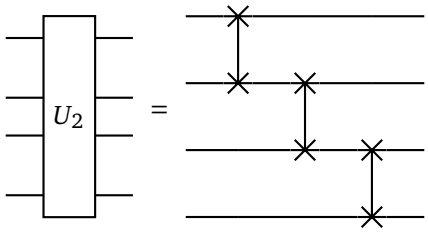
Theorem 8.10 (Modular powers · *cited*)**.** *For b coprime to $N < 2^n$, $\mathsf{C}(U_b)$ has an exact circuit of $O(n^2)$ gates over the alphabet, with a workspace of $O(n)$ qubits that starts and ends in $|0\cdots0\rangle$.*

The circuits are reversible as in Lecture 1 and exact as the Toffoli is ((6.2)); see Nielsen and Chuang, Box 5.2 and p. 232, Shor, pp. 10–11, and Watrous (2025), p. 221.

Proof of Theorem 8.4. What reaches the loop has $m \geq 2$ distinct primes, and a try finds a with probability at least $1 - 2^{-7}$. It then succeeds when Theorem 8.3 and Theorem 8.9 both do, with probability at least $\frac{1}{2} \cdot \frac{2}{3}$, by Proposition 8.2; three tries fail with probability at most $(1 - \frac{127}{384})^3 < 1/3$. A try runs $3n$ circuits of $O(n^3)$ gates: t controlled powers of $O(n^2)$ gates each, t Hadamards, and $\mathrm{QFT}_{2^t}^{-1}$ compiled within $1/200$ in $O(n^2 \log^c n)$ gates (Theorem 7.2). ∎

8.6 F15, in short

For $N = 15$ and $a = 2$ the orbit of 1 is 1, 2, 4, 8, so $r = 4$, and $2^2 = 4 \not\equiv -1$: $\gcd(3, 15) = 3$ and $\gcd(5, 15) = 5$. As $16 \equiv 1 \pmod{15}$, doubling a 4-bit y modulo 15 moves its top bit to the bottom, $y_1y_2y_3y_4 \mapsto y_2y_3y_4y_1$ ($y = 15$ is fixed either way):



(8.1)

an equality of matrices. So $U_4 = U_2^2$ is two swaps, $U_2^4 = \mathbb{1}$, and each controlled power is a few controlled swaps, exact. The phases $s/4$ are read exactly for every $t \geq 2$: $s = 1$ and $s = 3$ give $r = 4$, $s = 2$ only the divisor 2, which fails the test, and $s = 0$ nothing. An instance illustrates and proves nothing about scaling; $r = 4$ divides 2^t, and the order of 2 modulo 21 above, like Lecture 7's phase $\frac{1}{3}$, shows the general case.

For the small illustration $t = 3$, all possible readings and their classical processing are below. The denominator column lists distinct convergent denominators; a dash means that none passes.

y	$y/8$	$q < 15$	$2^q \equiv 1 \pmod{15}$
0	0/1	1	–
2	1/4	1, 4	4
4	1/2	1, 2	–
6	3/4	1, 4	4

The reading 4 estimates a reduced fraction with denominator 2, and $2^2 = 4 \not\equiv 1 \pmod{15}$, so that run retains nothing. The readings 2 and 6 retain 4. Across the runs the algorithm keeps the least retained value. In general a passing value can be a multiple of the order: verification alone does not certify minimality. The full algorithm uses $t = 2n + 1$, not this shortened illustration.

8.7 Status

Theorem 8.4 is about a function problem, and states an algorithm with its resources: bounded error, $O(n)$ circuits of $O(n^3)$ gates. It is proved modulo the cited lemmas. No classical algorithm of polynomial time is known; the best known, the number field sieve, runs in time $\exp(O(n^{1/3}(\log n)^{2/3}))$ by a heuristic analysis (Nielsen and Chuang, p. 216; de Wolf, p. 37). That none exists is a conjecture, on which much public-key cryptography rests. This is the evidence Lecture 6 promised for its conjecture BQP ≠ BPP: a quantum algorithm proved, a classical one not known. Unlike Simon's separation (Lecture 5), it needs no oracle, but no classical lower bound is proved.

A decision version. To compare with BQP and BPP, encode two integers $N \geq 2$ and $1 \leq k < N$ in binary, and ask whether a divisor lies below the bound. Define

$$D(N,k) = \begin{cases} 1, & \exists d : \ 1 < d \leq k, \ d \mid N, \\ 0, & \text{otherwise.} \end{cases}$$

Reject encodings outside this range. A complete factorisation decides this question: its least prime factor is the least divisor greater than one, and is compared with k. For a prime N the answer is zero.

The factor routine yields such a factorisation with bounded error, without assuming a new primality test. Execute its same bounded programme on an integer $M \geq 2$, accepting only verified proper factors. If a factor is found, split M into that factor and its quotient and process both. If none is found after K independent attempts, keep M as a leaf. A prime can never produce a verified proper factor. A composite is mistaken for a prime with probability at most 3^{-K}.

If the original N has n bits, there are at most n leaves, because each is at least two and their product is N. The splitting tree has fewer than $2n$ calls. Choose $K = \lceil \log_3(6n) \rceil$; by the union bound, the chance of any composite leaf is at most $2n \cdot 3^{-K} \leq 1/3$. The total quantum and classical work is polynomial. Verify candidates and retry; a majority vote on factors would be inappropriate, since different factors can all be correct.

Circuit composition · cited. Polynomial-time classical control between polynomially many uniformly generated quantum circuits can be converted to one polynomial-time generated circuit, with any chosen inverse-polynomial additional error. The conversion retains measurement records, implements classical processing reversibly and pads the calls to a fixed polynomial bound. In particular, the varying modulus and multiplier are inputs to uniform arithmetic circuits; exponentially many prewritten branch circuits are not required. This is the circuit composition used for factoring in Watrous, *An introduction to quantum information and quantum circuits* (2011), §§3.2 and 4.2, pp. 11–14.

Apply this cited fact to the bounded procedure above. First amplify its single decision bit to success at least 5/6 by Proposition 6.11, then allow at most 1/6 additional probability error in the conversion and compilation. The success remains at least 2/3. The resulting circuit starts in $|0 \cdots 0\rangle$: Hadamards prepare the coins, and an X prepares each required $|1\rangle$. Draw the final decision bit as its top wire and read it, as in Definition 6.9. Hence D is in BQP, using the cited factoring and circuit-composition ingredients. The vote here is on decision bits, not on factors.

Conversely, a correct solver for D first tests $D(N, N-1)$. A zero means that N is prime. Otherwise binary search for the first bound k with answer one finds the least proper divisor. Divide it out and repeat. There are polynomially many decision queries. A bounded-error solver's answers can first be amplified by Proposition 6.11 so that their total error remains

bounded. Hence an efficient classical bounded-error solver for this decision problem would also factor efficiently. This is the connection with decision classes; it supplies no classical lower bound.

Summary.

- *The reduction:* if r is even and $a^{r/2} \not\equiv -1$, $\gcd(a^{r/2} \pm 1, N)$ are proper factors (Proposition 8.2); a uniformly chosen unit qualifies with probability at least $\frac{1}{2}$ for odd N with at least two distinct prime factors, cited.
- *The phase:* $|1\rangle$ is the uniform superposition of the r orbit eigenvectors u_s of U_a, with the phases s/r (Lemma 8.5).
- *The main result:* $t = 2n + 1$, continued fractions, the coprimality bound and the least passing denominator give r with probability at least $2/3$ in $3n$ runs (Theorem 8.9), and a factor (Theorem 8.4).
- *Status:* proved modulo cited lemmas; classical hardness is conjectured.
- *Next:* Lecture 9, search.

References

Nielsen and Chuang, §5.3, pp. 226–235, Box 5.2, p. 228, Theorem A4.16, pp. 637–638, and p. 216. Kitaev, Shen and Vyalyi, §4.2, pp. 38–41, §13.3, pp. 120–122, and Theorem A.13, p. 250. Watrous (2025), pp. 218–228. De Wolf, pp. 37–38. P. W. Shor, *Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer*, arXiv:quant-ph/9508027 (1995), pp. 8–16.

J. Watrous, *An introduction to quantum information and quantum circuits* (2011), §§3.2 and 4.2, pp. 11–14.